Final Consulting Project Report

Preschool Expulsion and Suspension

Anurag Banerjee
STAT 5760
Ohio State University

June 22, 2026

Client:
Semester: Spring 2026

Contents

1 Summary

This project supported in studying the relationship between language development and preschool expulsion risk. To help prepare for the eventual study, I conducted a statistical analysis on a simulated dataset designed to resemble the structure of the data expected in the full project. The simulated data included demographic variables such as sex, race, and household income, along with multiple assessment scores measuring language, behavior, cognitive communication, and expulsion related risk indicators.

The analysis had two main goals. First, it aimed to determine whether children classified as at risk for expulsion differed from children not at risk in language, behavior, cognitive communication, and PERM-related measures. Second, it assessed whether adding behavior to a model that already included language explained significantly additional variability. Since many of the language and cognitive variables were expected to be correlated within domain, the analysis emphasized identifying underlying latent factors rather than treating every test score as fully independent.

2 Project Objectives

2.1 Problem Statement

The project focused on two related statistical problems. First, the client wanted to determine whether children classified as at risk for expulsion differed from children not at risk on measures of language, behavior, cognitive communication, and PERM score. Second, the client wanted to assess whether behavior contributed meaningful additional explanatory value in a predictive model that already included language. A key complication was that many of the assessment variables within the same domain, especially language, were highly correlated and likely reflected underlying latent constructs rather than distinct independent traits. As a result, the problem was not only to compare groups and evaluate predictive relationships, but also to reduce collinearity and identify interpretable latent factors that could be used in a more stable and meaningful statistical analysis.

2.2 Project Objectives

The objectives of this project were:

- Do the at-risk and not at-risk kids differ significantly in terms of behavior, language, conginitive communication and expulsion-risk related measures?
- Does behavior contribute significant additional explained variability to a model with only language?

3 Deliverables

The project deliverables included a written report, reproducible analysis code, and supporting explanation of the statistical methods so that the client could understand, replicate, and extend the work. In addition to producing the formal analyses, the project emphasized clear communication of interpreting and testing models, MANOVA and factor-analytic methods in a way that supported practical decision-making and future use by the client.

4 Data

The analysis used a simulated dataset created by the client to approximate the structure of the data expected in the planned study of preschool expulsion risk. Each observation represented a child and included demographic stratification variables, a binary indicator of expulsion-risk status, and a set of assessment scores intended to capture multiple developmental domains relevant to the research questions.

The demographic variables included sex, race, and household income. The outcome-related grouping variable identified whether a child was classified as at risk or not at risk for expulsion. In addition, the dataset contained multiple observed test scores spanning four substantive domains. Language-related measures included `CASL_SentExp`, `CASL_Rec`, `CASL_Prag`, `CASL_Exp`, `CASL_GLAI`, `Vineland_Exp`, and `Vineland_Rec`. Behavioral functioning was represented by `SDQ_Total`, cognitive communication was represented by `BRIEF_P_T` and `HTKS_Score`, and `PERM_Score` was retained as a direct measure related to expulsion risk.

Because several variables within the same domain were designed to measure related aspects of a broader construct, the dataset was expected to exhibit substantial within-domain correlation, particularly among the language measures. This structure made the data suitable for both direct group comparisons and latent-variable methods aimed at reducing collinearity and summarizing the main constructs underlying the observed assessment scores.

5 Methods

The analysis addressed two objectives related to preschool expulsion risk using the client's simulated dataset. Children were classified into two groups, at risk and not at risk of expulsion, and this grouping was used throughout the analysis as the main comparison of interest.

The analysis began with descriptive summaries, correlation analysis, and exploratory comparisons between the at-risk and not-at-risk groups. Correlation plots and bivariate summaries were used to understand patterns of dependence both within and across domains. Variance inflation factors were also examined to assess collinearity among the observed variables and to motivate the use of latent-variable methods.

For an initial group-comparison analysis, representative observed variables were selected for each domain. Language was represented by `CASL_GLAI`, cognitive communication was summarized using the average of `BRIEF_P_T` and `HTKS_Score`, behavior was represented by `SDQ_Total`, and expulsion-related functioning was represented by `PERM_Score`. These variables were then analyzed using MANOVA to determine whether the at-risk and not-at-risk groups differed in their multivariate profile. Linear discriminant analysis and individual mean comparisons were used as supporting tools to understand which domains appeared to be contributing most strongly to any group differences.

Because the language measures were strongly correlated and appeared to reflect more than one underlying construct, factor analysis was then used to extract latent factors for language. A separate factor analysis was also used to extract one latent factor for cognitive communication from `BRIEF_P_T` and `HTKS_Score`.

After these latent variables were constructed, the main multivariate group-comparison analysis was repeated using the reduced feature set. This provided a more interpretable and statistically stable analysis than relying only on individual observed test scores.

The second objective was evaluated using nested logistic regression models for expulsion-risk group membership. A reduced model containing language latent factors was compared against a larger model that additionally included `SDQ_Total`. These models were compared using a likelihood-ratio chi-square test to assess whether behavior explained additional variability beyond language alone.

6 Findings

The preliminary descriptive and correlational analyses showed substantial dependence among the observed variables, especially within the language domain. In particular, `CASL_SentExp` and `CASL_Prag` were almost perfectly correlated and appeared to move together in a way that was distinct from the remaining language measures. The cognitive communication measures, `BRIEF_P_T` and `HTKS_Score`, were not identical but showed similar relationships with other variables, suggesting that they were likely reflecting a common underlying construct.

In the initial analysis using representative observed variables, the MANOVA indicated evidence of an overall difference between the at-risk and not-at-risk groups. Follow-up analyses suggested that behavior contributed strongly to this separation, while the univariate results were less consistent once multiplicity was considered. This initial stage nevertheless indicated that the two groups differed enough to justify a more refined latent-variable analysis.

The factor analysis of the language variables provided one of the clearest findings in the project. Parallel analysis supported a two-factor solution, and the resulting loading structure aligned well with the substantive interpretation of the measures. One latent factor represented the broader set of general language measures, while the second latent factor was driven primarily by sentence expression and pragmatics. This was an important result because it showed that the language domain was not adequately summarized by a single observed score alone and that sentence expression and pragmatics formed a particularly meaningful dimension for distinguishing children by expulsion-risk status.

A single latent factor was also extracted for cognitive communication. This supported the idea that `BRIEF_P_T` and `HTKS_Score` were capturing similar underlying information.

Using the latent-factor representation, the MANOVA again showed a strong overall difference between the at-risk and not-at-risk groups. The follow-up analyses indicated that the second language latent factor, corresponding primarily to sentence expression and pragmatics, was especially important in separating the two groups. In addition, `SDQ_Total` and `PERM_Score` also showed evidence of mean differences between groups. By contrast, the first language latent factor and the cognitive communication latent factor were less central in distinguishing at-risk from not-at-risk children in the final analysis.

The linear discriminant analysis supported the same overall interpretation. The dominant direction of group separation was driven most strongly by the second language latent factor, with smaller contributions from the remaining variables. This reinforced the conclusion that the sentence expression and pragmatics dimension of language was one of the most informative features in the simulated dataset for distinguishing expulsion-risk status.

The nested logistic regression analysis addressed the second project objective by testing whether behavior improved a model that already included language. When the reduced model containing only the two language latent factors was compared with the larger model that also included `SDQ_Total`, the likelihood-ratio test did not provide sufficient evidence that behavior explained significant additional variability beyond language. Under this simulated-data setup, the main predictive signal appeared to be captured primarily through the language structure rather than through incremental contribution from behavior once language was already included.

Overall, the findings suggest that children classified as at risk and not at risk of expulsion differed meaningfully across the measured domains, with the clearest evidence arising from the language dimension associated with sentence expression and pragmatics. Behavior and `PERM_Score` also appeared relevant in group comparisons, but the nested-model analysis did not show enough evidence that behavior improved prediction after accounting for language. Because these results were obtained from simulated data and some of the supporting univariate tests were interpreted without a formal multiple-testing correction in the final stage, the findings should be viewed as a proof of concept for the client's planned study design rather than as definitive scientific conclusions.

7 Future Improvements

The most important next step is to repeat the full analysis using real study data once data collection is complete. Because the current project relied on simulated data, the latent structure identified in the language variables, especially the interpretation of the factor associated with sentence expression and pragmatics, should be re-evaluated on the observed dataset. Once real data are available, the factor analysis, multivariate group comparisons, and predictive modeling steps should all be rerun to assess whether the same substantive patterns remain supported.

A second improvement concerns the handling of multiple testing in the follow-up analyses. While MANOVA provides an overall multivariate test, the subsequent individual comparisons and supporting tests can inflate the Type I error rate when many hypotheses are examined at once. Future versions of the analysis should therefore incorporate an appropriate multiple-testing adjustment when moving beyond the overall multivariate test, so that any domain-specific findings are interpreted with stronger control of false positive results.

A third area for future development is formal model comparison across different subsets of predictors. The client indicated interest in comparing several candidate models that use different combinations of language, behavior, cognitive communication, and related features, with the goal of determining which model is most strongly supported and by what margin. This objective is naturally suited to a Bayesian modeling framework, where posterior probabilities can be used to compare competing models in a probabilistic way. Under such an approach, multiple candidate logistic-regression models could be fit and evaluated using posterior model probabilities, allowing the client to quantify uncertainty in model choice and directly compare the relative support for alternative predictor sets.